Assignment 1 for Cloud Computing

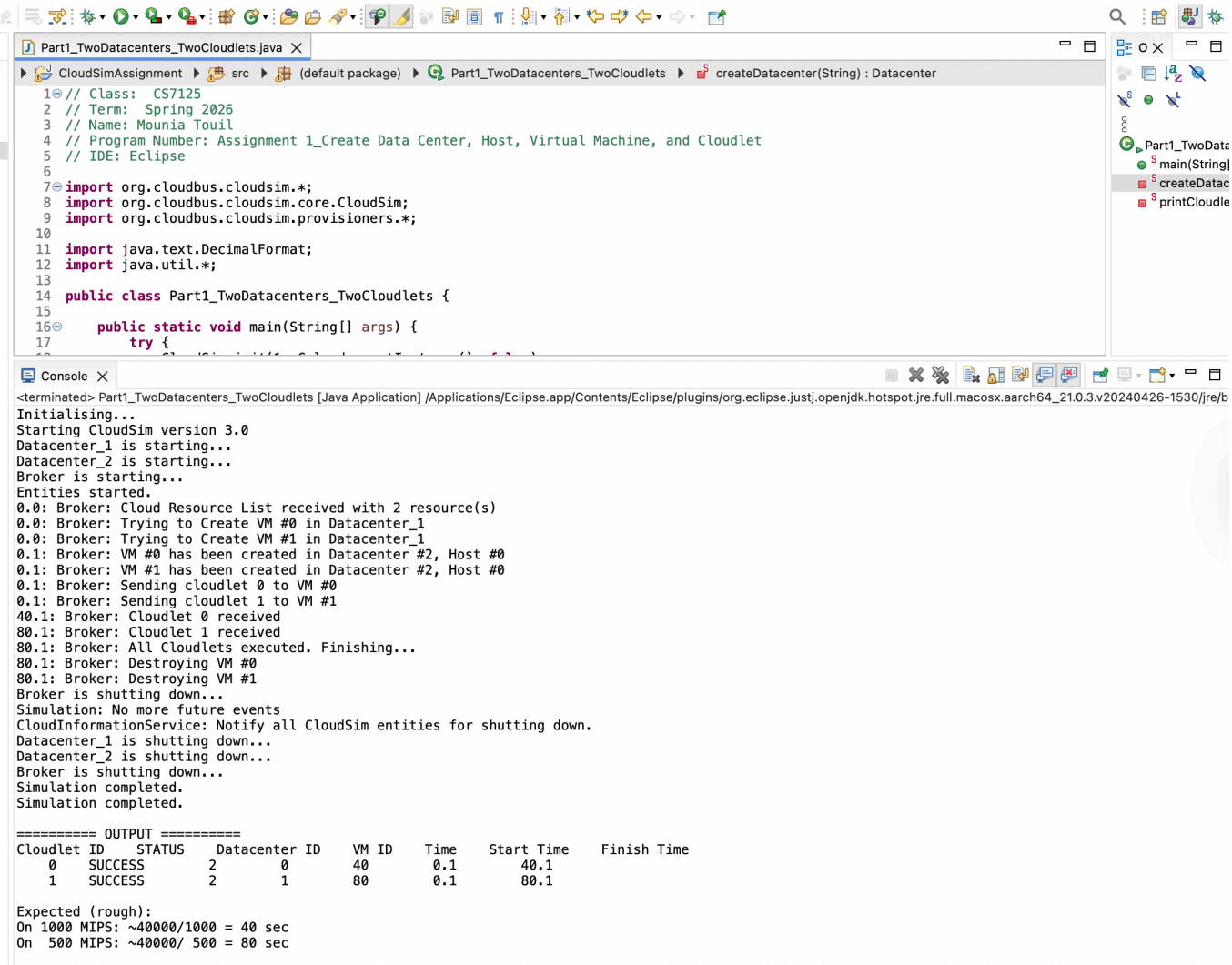
Mounia Touil

Create Data Center, Host, Virtual Machine, and Cloudlet.

Screenshots of the programs running:

Part 1 (Program: Part1_TwoDatacenters_TwoCloudlets.java)

This program runs two cloudlets on two virtual machines that have different MIPS values. The cloudlet on the faster virtual machine finishes earlier, showing how processing power affects execution time.



```
Part1_TwoDatacenters_TwoCloudlets.java
CloudSimAssignment ▸ src ▸ (default package) ▸ Part1_TwoDatacenters_TwoCloudlets ▸ createDatacenter(String) : Datacenter

1  // Class:  CS7125
2  // Term:  Spring 2026
3  // Name: Mounia Touil
4  // Program Number: Assignment 1_Create Data Center, Host, Virtual Machine, and Cloudlet
5  // IDE: Eclipse
6
7  import org.cloudbus.cloudsim.*;
8  import org.cloudbus.cloudsim.core.CloudSim;
9  import org.cloudbus.cloudsim.provisioners.*;
10
11 import java.text.DecimalFormat;
12 import java.util.*;
13
14 public class Part1_TwoDatacenters_TwoCloudlets {
15
16     public static void main(String[] args) {
17         try {

Part1_TwoData
main(String|
createDatac
printCloudle

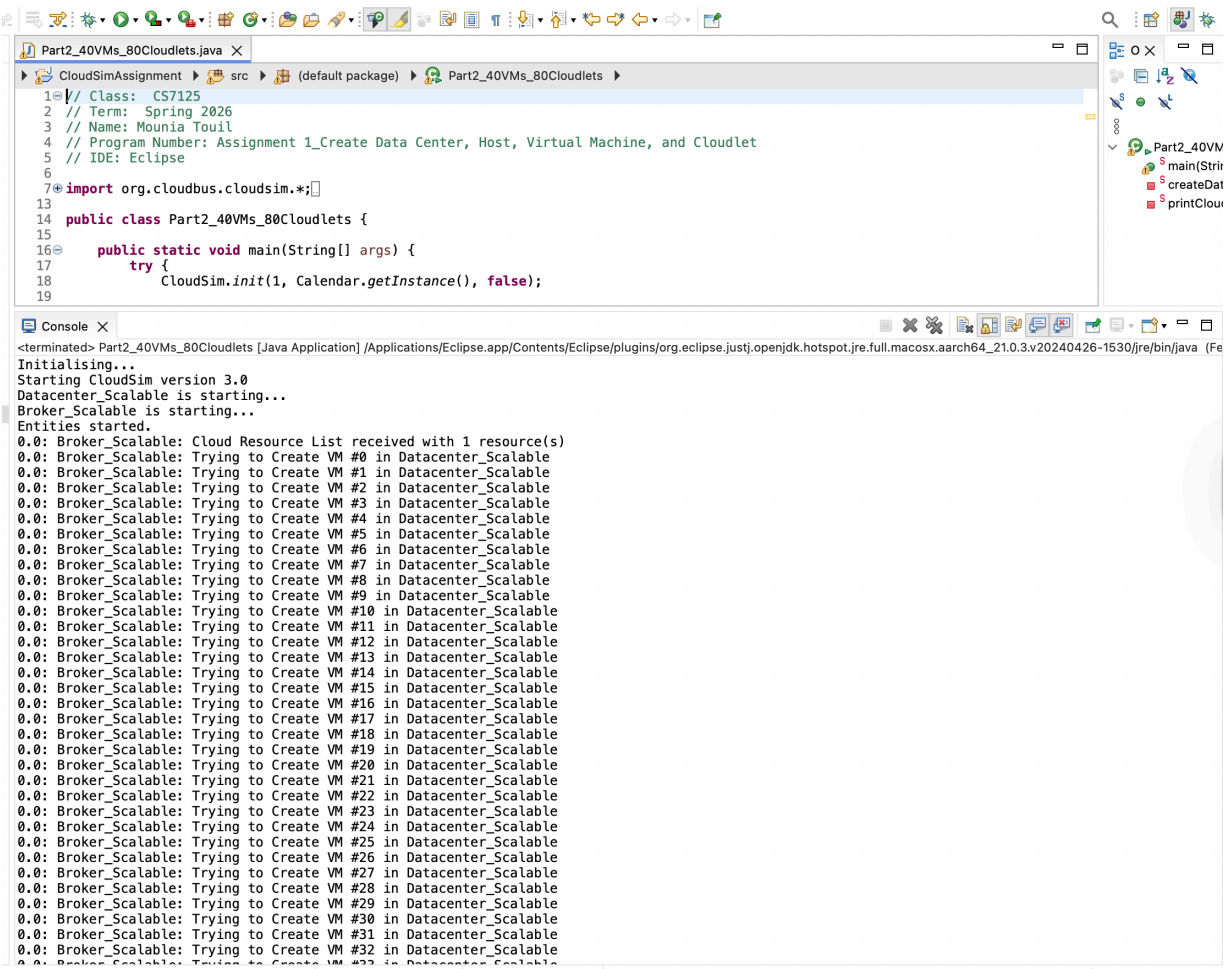
Console
<terminated> Part1_TwoDatacenters_TwoCloudlets [Java Application] /Applications/Eclipse.app/Contents/Eclipse/plugins/org.eclipse.justj.openjdk.hotspot.jre.full.macosx.aarch64_21.0.3.v20240426-1530/jre/b
Initialising...
Starting CloudSim version 3.0
Datacenter_1 is starting...
Datacenter_2 is starting...
Broker is starting...
Entities started.
0.0: Broker: Cloud Resource List received with 2 resource(s)
0.0: Broker: Trying to Create VM #0 in Datacenter_1
0.0: Broker: Trying to Create VM #1 in Datacenter_1
0.1: Broker: VM #0 has been created in Datacenter #2, Host #0
0.1: Broker: VM #1 has been created in Datacenter #2, Host #0
0.1: Broker: Sending cloudlet 0 to VM #0
0.1: Broker: Sending cloudlet 1 to VM #1
40.1: Broker: Cloudlet 0 received
80.1: Broker: Cloudlet 1 received
80.1: Broker: All Cloudlets executed. Finishing...
80.1: Broker: Destroying VM #0
80.1: Broker: Destroying VM #1
Broker is shutting down...
Simulation: No more future events
CloudInformationService: Notify all CloudSim entities for shutting down.
Datacenter_1 is shutting down...
Datacenter_2 is shutting down...
Broker is shutting down...
Simulation completed.
Simulation completed.

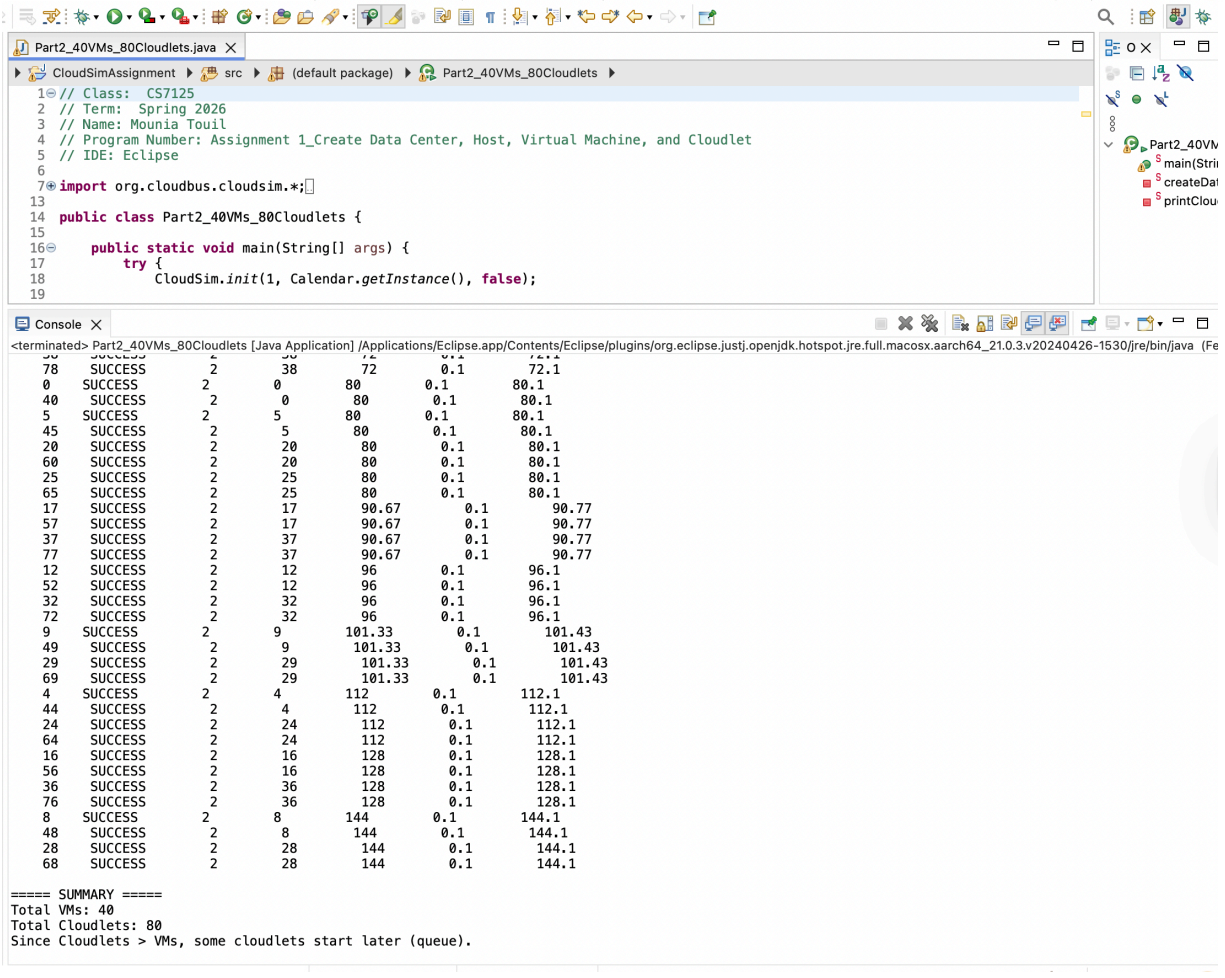
========== OUTPUT ==========
Cloudlet ID    STATUS    Datacenter ID    VM ID    Time    Start Time    Finish Time
    0    SUCCESS        2        0        40        0.1        40.1
    1    SUCCESS        2        1        80        0.1        80.1

Expected (rough):
On 1000 MIPS: ~40000/1000 = 40 sec
On  500 MIPS: ~40000/ 500 = 80 sec
```

Part 2 (Program: Part2_40VMs_80Cloudlets.java)

This program runs 80 cloudlets using 40 virtual machines in a single datacenter. Since there are more cloudlets than virtual machines, some cloudlets have to wait and start later, which shows how scheduling and scalability work in the cloud.


Part2_40VMs_80Cloudlets.java
CloudSimAssignment ▸ src ▸ (default package) ▸ Part2_40VMs_80Cloudlets
1 // Class: CS7125
2 // Term: Spring 2026
3 // Name: Mounia Touil
4 // Program Number: Assignment 1_Create Data Center, Host, Virtual Machine, and Cloudlet
5 // IDE: Eclipse
6
7 import org.cloudbus.cloudsim.*;
13
14 public class Part2_40VMs_80Cloudlets {
15
16 public static void main(String[] args) {
17 try {
18 CloudSim.init(1, Calendar.getInstance(), false);
19
Console
<terminated> Part2_40VMs_80Cloudlets [Java Application] /Applications/Eclipse.app/Contents/Eclipse/plugins/org.eclipse.justj.openjdk.hotspot.jre.full.macosx.aarch64_21.0.3.v20240426-1530/jre/bin/java
Initialising...
Starting CloudSim version 3.0
Datacenter_Scalable is starting...
Broker_Scalable is starting...
Entities started.
0.0: Broker_Scalable: Cloud Resource List received with 1 resource(s)
0.0: Broker_Scalable: Trying to Create VM #0 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #1 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #2 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #3 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #4 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #5 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #6 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #7 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #8 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #9 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #10 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #11 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #12 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #13 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #14 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #15 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #16 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #17 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #18 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #19 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #20 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #21 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #22 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #23 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #24 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #25 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #26 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #27 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #28 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #29 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #30 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #31 in Datacenter_Scalable
0.0: Broker_Scalable: Trying to Create VM #32 in Datacenter_Scalable


Part2_40VMs_80Cloudlets.java
CloudSimAssignment ▸ src ▸ (default package) ▸ Part2_40VMs_80Cloudlets
1 // Class: CS7125
2 // Term: Spring 2026
3 // Name: Mounia Touil
4 // Program Number: Assignment 1_Create Data Center, Host, Virtual Machine, and Cloudlet
5 // IDE: Eclipse
6
7 import org.cloudbus.cloudsim.*;
13
14 public class Part2_40VMs_80Cloudlets {
15
16 public static void main(String[] args) {
17 try {
18 CloudSim.init(1, Calendar.getInstance(), false);
19
Console
<terminated> Part2_40VMs_80Cloudlets [Java Application] /Applications/Eclipse.app/Contents/Eclipse/plugins/org.eclipse.justj.openjdk.hotspot.jre.full.macosx.aarch64_21.0.3.v20240426-1530/jre/bin/java
78 SUCCESS 2 38 72 0.1 72.1
0 SUCCESS 2 0 80 0.1 80.1
40 SUCCESS 2 0 80 0.1 80.1
5 SUCCESS 2 5 80 0.1 80.1
45 SUCCESS 2 5 80 0.1 80.1
20 SUCCESS 2 20 80 0.1 80.1
60 SUCCESS 2 20 80 0.1 80.1
25 SUCCESS 2 25 80 0.1 80.1
65 SUCCESS 2 25 80 0.1 80.1
17 SUCCESS 2 17 90.67 0.1 90.77
57 SUCCESS 2 17 90.67 0.1 90.77
37 SUCCESS 2 37 90.67 0.1 90.77
77 SUCCESS 2 37 90.67 0.1 90.77
12 SUCCESS 2 12 96 0.1 96.1
52 SUCCESS 2 12 96 0.1 96.1
32 SUCCESS 2 32 96 0.1 96.1
72 SUCCESS 2 32 96 0.1 96.1
9 SUCCESS 2 9 101.33 0.1 101.43
49 SUCCESS 2 9 101.33 0.1 101.43
29 SUCCESS 2 29 101.33 0.1 101.43
69 SUCCESS 2 29 101.33 0.1 101.43
4 SUCCESS 2 4 112 0.1 112.1
44 SUCCESS 2 4 112 0.1 112.1
24 SUCCESS 2 24 112 0.1 112.1
64 SUCCESS 2 24 112 0.1 112.1
16 SUCCESS 2 16 128 0.1 128.1
56 SUCCESS 2 16 128 0.1 128.1
36 SUCCESS 2 36 128 0.1 128.1
76 SUCCESS 2 36 128 0.1 128.1
8 SUCCESS 2 8 144 0.1 144.1
48 SUCCESS 2 8 144 0.1 144.1
28 SUCCESS 2 28 144 0.1 144.1
68 SUCCESS 2 28 144 0.1 144.1
===== SUMMARY =====
Total VMs: 40
Total Cloudlets: 80
Since Cloudlets > VMs, some cloudlets start later (queue).

Complete Running Results:

Output for Part 1 (Program: Part1_TwoDatacenters_TwoCloudlets.java):

```
========== OUTPUT ==========
Cloudlet ID    STATUS    Datacenter ID    VM ID    Time    Start Time
Finish Time
    0    SUCCESS        2        0        40        0.1        40.1
    1    SUCCESS        2        1        80        0.1        80.1

Expected (rough):
On 1000 MIPS: ~40000/1000 = 40 sec
On  500 MIPS: ~40000/ 500 = 80 sec
```

After running Program 1, the console output shows that CloudSim initializes successfully and starts two data centers named Datacenter_1 and Datacenter_2. Each data center contains one host. A broker is created to manage the virtual machines and cloudlets. Two virtual machines broker then requests two virtual machines, and both are created successfully because sufficient resources are available in the data centers.

Each cloudlet is assigned to a different virtual machine because both VMs are available and idle. The start time for both cloudlets is almost the same, which shows there is no waiting or queueing. The cloudlet running on the virtual machine with higher MIPS finishes earlier, while the cloudlet on the lower MIPS virtual machine finishes later. This difference happens because MIPS represents how fast a virtual machine can process instructions, so a higher MIPS results in a shorter execution time.
The virtual machines are created in the data centers that have available hosts and enough resources. The broker does not choose a random datacenter; it selects one that can satisfy the VM requirements. Similarly, cloudlets are sent to virtual machines that are available at the time of scheduling, which is why each cloudlet is assigned to a specific VM.

Output for Part 2 (Program: Part2_40VMs_80Cloudlets.java):

```
========== OUTPUT ==========
Cloudlet ID    STATUS    Datacenter ID    VM ID    Time    Start Time    Finish Time
    11    SUCCESS        2        11        35.2        0.1        35.3
    51    SUCCESS        2        11        35.2        0.1        35.3
    31    SUCCESS        2        31        35.2        0.1        35.3
    71    SUCCESS        2        31        35.2        0.1        35.3
    10    SUCCESS        2        10        40        0.1        40.1
    50    SUCCESS        2        10        40        0.1        40.1
    30    SUCCESS        2        30        40        0.1        40.1
    70    SUCCESS        2        30        40        0.1        40.1
    3    SUCCESS        2        3        41.6        0.1        41.7
    43    SUCCESS        2        3        41.6        0.1        41.7
    23    SUCCESS        2        23        41.6        0.1        41.7
    63    SUCCESS        2        23        41.6        0.1        41.7
    15    SUCCESS        2        15        48        0.1        48.1
    55    SUCCESS        2        15        48        0.1        48.1
    35    SUCCESS        2        35        48        0.1        48.1
    75    SUCCESS        2        35        48        0.1        48.1
    2    SUCCESS        2        2        48        0.1        48.1
    42    SUCCESS        2        2        48        0.1        48.1
    22    SUCCESS        2        22        48        0.1        48.1
    62    SUCCESS        2        22        48        0.1        48.1
    7    SUCCESS        2        7        54.4        0.1        54.5
    47    SUCCESS        2        7        54.4        0.1        54.5
    27    SUCCESS        2        27        54.4        0.1        54.5
    67    SUCCESS        2        27        54.4        0.1        54.5
    14    SUCCESS        2        14        56        0.1        56.1
    54    SUCCESS        2        14        56        0.1        56.1
    34    SUCCESS        2        34        56        0.1        56.1
    74    SUCCESS        2        34        56        0.1        56.1
    1    SUCCESS        2        1        58.67        0.1        58.77
    41    SUCCESS        2        1        58.67        0.1        58.77
    21    SUCCESS        2        21        58.67        0.1        58.77
    61    SUCCESS        2        21        58.67        0.1        58.77
    19    SUCCESS        2        19        60.8        0.1        60.9
    59    SUCCESS        2        19        60.8        0.1        60.9
    39    SUCCESS        2        39        60.8        0.1        60.9
    79    SUCCESS        2        39        60.8        0.1        60.9
    6    SUCCESS        2        6        64        0.1        64.1
    46    SUCCESS        2        6        64        0.1        64.1
    26    SUCCESS        2        26        64        0.1        64.1
    66    SUCCESS        2        26        64        0.1        64.1
    13    SUCCESS        2        13        69.33        0.1        69.43
    53    SUCCESS        2        13        69.33        0.1        69.43
    33    SUCCESS        2        33        69.33        0.1        69.43
    73    SUCCESS        2        33        69.33        0.1        69.43
    18    SUCCESS        2        18        72        0.1        72.1
    58    SUCCESS        2        18        72        0.1        72.1
    38    SUCCESS        2        38        72        0.1        72.1
    78    SUCCESS        2        38        72        0.1        72.1
    0    SUCCESS        2        0        80        0.1        80.1
    40    SUCCESS        2        0        80        0.1        80.1
    5    SUCCESS        2        5        80        0.1        80.1
    45    SUCCESS        2        5        80        0.1        80.1
    20    SUCCESS        2        20        80        0.1        80.1
    60    SUCCESS        2        20        80        0.1        80.1
    25    SUCCESS        2        25        80        0.1        80.1
    65    SUCCESS        2        25        80        0.1        80.1
    17    SUCCESS        2        17        90.67        0.1        90.77
    57    SUCCESS        2        17        90.67        0.1        90.77
    37    SUCCESS        2        37        90.67        0.1        90.77
    77    SUCCESS        2        37        90.67        0.1        90.77
    12    SUCCESS        2        12        96        0.1        96.1
    52    SUCCESS        2        12        96        0.1        96.1
    32    SUCCESS        2        32        96        0.1        96.1
    72    SUCCESS        2        32        96        0.1        96.1
    9    SUCCESS        2        9        101.33        0.1        101.43
    49    SUCCESS        2        9        101.33        0.1        101.43
    29    SUCCESS        2        29        101.33        0.1        101.43
    69    SUCCESS        2        29        101.33        0.1        101.43
    4    SUCCESS        2        4        112        0.1        112.1
    44    SUCCESS        2        4        112        0.1        112.1
    24    SUCCESS        2        24        112        0.1        112.1
    64    SUCCESS        2        24        112        0.1        112.1
    16    SUCCESS        2        16        128        0.1        128.1
    56    SUCCESS        2        16        128        0.1        128.1
    36    SUCCESS        2        36        128        0.1        128.1
    76    SUCCESS        2        36        128        0.1        128.1
    8    SUCCESS        2        8        144        0.1        144.1
    48    SUCCESS        2        8        144        0.1        144.1
    28    SUCCESS        2        28        144        0.1        144.1
    68    SUCCESS        2        28        144        0.1        144.1

===== SUMMARY =====
Total VMs: 40
Total Cloudlets: 80
Since Cloudlets > VMs, some cloudlets start later (queue).
```

In Program 2, the simulation begins with the initialization of a single datacenter configured with multiple physical hosts. The broker then initiates the creation of 40 virtual machines (VMs). The success of this phase is determined by the host's physical capacity, specifically the availability of RAM, MIPS, and bandwidth. Because the combined resources of the multiple hosts meet the aggregate demand, all 40 VMs are successfully provisioned.

Following VM creation, 80 cloudlets are submitted to the broker for processing. Since the number of cloudlets (80) exceeds the number of available VMs (40), a resource bottleneck occurs. Initially, only 40 cloudlets can execute concurrently; the remaining 40 tasks are placed in a wait queue. This queuing mechanism introduces a delay, which is reflected in the increased start times observed in the console output for the second batch of cloudlets.

The final execution logs show that cloudlets finish at different intervals for two primary reasons:

Staggered Start Times: Cloudlets that remained in the queue necessarily finish later than those that started immediately.

Heterogeneous VM Performance: The VMs vary in MIPS (Millions of Instructions Per Second) capacity. Tasks assigned to high-performance VMs complete faster than those on lower-tier VMs.

Furthermore, cloudlet-to-VM mapping is not random; the broker implements a specific allocation policy based on availability.